```vhdl
library ieee;
use ieee.std_logic_1164.all;

entity sixteen_to_eightEncoder is
   port(input3, input2, input1, input0: in std_logic_vector(3 downto 0);
        s:     in std_logic_vector(3 downto 0);
        en:    in std_logic;
        output16_8En1, output16_8En0: out std_logic_vector(3 downto 0));
end;

architecture synth of sixteen_to_eightEncoder is
begin
   process (s(3), s(2)) begin
      if (s(3) = '1' and s(2) = '1' and en = '1') then
         output16_8En1(3 downto 0) <= input3(3 downto 0);
      elsif (s(3) = '1' and s(2) = '0' and en = '1') then
         output16_8En1(3 downto 0) <= input2(3 downto 0);
      elsif (s(3) = '0' and s(2) = '1' and en = '1') then
         output16_8En1(3 downto 0) <= input1(3 downto 0);
      elsif (s(3) = '0' and s(2) = '0' and en = '1') then
         output16_8En1(3 downto 0) <= input0(3 downto 0);
      end if;
   end process;

   process (s(1), s(0)) begin
      if (s(1) = '1' and s(0) = '1' and en = '1') then
         output16_8En0(3 downto 0) <= input3(3 downto 0);
      elsif (s(1) = '1' and s(0) = '0' and en = '1') then
         output16_8En0(3 downto 0) <= input2(3 downto 0);
      elsif (s(1) = '0' and s(0) = '1' and en = '1') then
         output16_8En0(3 downto 0) <= input1(3 downto 0);
      elsif (s(1) = '0' and s(0) = '0' and en = '1') then
         output16_8En0(3 downto 0) <= input0(3 downto 0);
      end if;
   end process;
end;
```